rules for diagrams with *all* external lines off the mass shell, and then obtain S-matrix elements by letting the momenta associated with external lines approach their appropriate mass shells.

Feynman graphs with lines off the mass shell are just a special case of a wider generalization of the Feynman rules that takes into account the effects of various possible external fields. Suppose we add a sum of terms involving external fields $\epsilon_a(x)$ to the Hamiltonian, so that the interaction $V(t)$ that is used in the Dyson series (3.5.10) for the S-matrix is replaced with

$$V_\epsilon(t) = V(t) + \sum_a \int d^3x\, \epsilon_a(\mathbf{x}, t)\, o_a(\mathbf{x}, t) \,. \tag{6.4.1}$$

The 'currents' $o_a(t)$ have the usual time-dependence of the interaction picture

$$o_a(t) = \exp(iH_0 t) o_a(0) \exp(-iH_0 t) \,, \tag{6.4.2}$$

but are otherwise quite arbitrary operators. The S-matrix for any given transition $\alpha \to \beta$ then becomes a functional $S_{\beta\alpha}[\epsilon]$ of the c-number functions $\epsilon_a(t)$. The Feynman rules for computing this functional are given by an obvious extension of the usual Feynman rules. In addition to the usual vertices obtained from $V(t)$, we must include additional vertices: if $o_a(x)$ is a product of n_a field factors, then any o_a vertex with position label x must have n_a lines of corresponding types attached, and makes a contribution to the position-space Feynman rules equal to $-i\epsilon_a(x)$ times whatever numerical factors appear in $o_a(x)$. It follows then that the rth variational derivative of $S_{\beta\alpha}[\epsilon]$ with respect to $\epsilon_a(x)$, $\epsilon_b(y) \cdots$ at $\epsilon = 0$ is given by position space diagrams with r additional vertices, to which are attached respectively $n_a, n_b \cdots$ internal lines, and no external lines. These vertices carry position labels $x, y \cdots$ over which we do *not* integrate; each such vertex makes a contribution equal to $-i$ times whatever numerical factors appear in the associated current o_a.

In particular, in the case where these currents are all single field factors, i.e.,

$$V_\epsilon(t) = V(t) + \sum_\ell \int d^3x\, \epsilon_\ell(\mathbf{x}, t)\, \psi_\ell(\mathbf{x}, t) \,,$$

the rth variational derivative of $S_{\beta\alpha}[\epsilon]$ with respect to $\epsilon_\ell(x)$, $\epsilon_m(y) \cdots$ at $\epsilon = 0$ is given by position space diagrams with r additional vertices carrying spacetime labels $x, y \cdots$, to each of which is attached a single internal particle line of type $\ell, m \cdots$. These can be thought of as off-shell external lines, with the difference that their contribution to the matrix element is not a coefficient function like $(2\pi)^{-3/2} u_\ell(\mathbf{p}, \sigma) e^{ip\cdot x}$ or $(2\pi)^{-3/2} u_\ell^*(\mathbf{p}, \sigma) e^{-ip\cdot x}$ but a *propagator*, as well as a factor $-i$ from the vertex at the end of